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SPECIAL SECTION ARTICLE

Applications of artificial intelligence in pancreatic and biliary diseases

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Abstract

The application of artificial intelligence (AI) in medicine has increased rapidly with respect to tasks including disease detection/diagnosis, risk stratification, and prognosis prediction. With recent advances in computing power and algorithms, AI has shown promise in taking advantage of vast electronic health data and imaging studies to supplement clinicians. Machine learning and deep learning are the most widely used AI methodologies for medical research and have been applied in pancreatobiliary diseases for which diagnosis and treatment selection are often complicated and require joint consideration of data from multiple sources. The aim of this review is to provide a concise introduction of the major AI methodologies and the current landscape of AI research in pancreatobiliary diseases.

Introduction

Artificial Intelligence (AI) refers to the technology of using computers to simulate behaviors and thinking processes equivalent to the human being. With the availability of big data and progress in computing capabilities, there is a rapid increase in applying AI in medical research.[1] The applications of AI in medicine can be broadly categorized based on the types of data into numerical or image-based inputs. Application of AI in medical image analysis has emerged as the most promising direction because of the high performance in the identification of imaging abnormalities, lesion detection, and characterization.[2] It provides a new tool that assists clinicians in a variety of diagnostic tasks and supplements the unmet need in various clinical scenarios.

Two main classes of AI methods are used in medical image analysis. Machine learning algorithms calculate human-crafted features based on domain knowledge and then identify the best combinations of features to classify lesions and support clinical decision-making.[3] Frequently coupled with machine learning algorithms is radiomics, a widely used tool that enables high-throughput extraction of features representing quantitative information on density, shape, and texture from the images.[4] By contrast, deep learning (DL) algorithms learn automatically from images to identify the significant features without requiring human-crafted features and show promise to detect subtle imaging characteristics that are imperceptible to the naked eye.[5]

The diagnosis, evaluation, and treatment planning of pancreatobiliary diseases are complex and rely heavily on medical imaging. The application of AI for pancreatobiliary diseases is increasing, mainly for image analysis with respect to tumor detection, lesion characterization, and prognosis prediction. In addition, non-imaging data such as numerical or categorical clinical variables can also be analyzed using AI methods, either alone or in conjunction with images. The aim of this review is to summarize the current landscape of applying AI in pancreatobiliary diseases and provide a concise review of the clinical needs addressed and the performance achieved with AI methodologies in current literature (Table 1). The application of AI in diabetes mellitus is not included in the scope of this review.

Radiomics and machine learning

Radiomics refers to the extraction of quantitative features from the images for applications such as detection or diagnosis. The analyzed images can be obtained from computed tomography (CT), magnetic resonance imaging (MRI), or positron emission tomography (PET). The radiomic features are designed to capture various forms of imaging characteristics in the images including intensity, shape, and texture and are subsequently used to predict outcomes of interest (e.g. presence of disease or prognosis).[6] Extracted radiomic features can be analyzed with traditional biostatistical methods such as a logistic regression model or with machine learning methods such as support vector machine (SVM)[7] and random forests.[8] Logistic regression is widely used for binary classification (e.g. with or without disease). SVM enables classification by finding a model that separates observations (e.g. patients) of different classes (e.g. with or without disease) as widely as possible (i.e. finding a plane in the high-dimensional space of the radiomic features that has the greatest distance to each class). By contrast, random forest performs classification by first constructing multiple simple decision trees, each with a randomly selected subset of data (e.g. patients) and features. The final model predicts

Table 1 Summary of reviewed literature

Author	Year	Clinical needs addressed	Data type	Case number	Sensitivity/specificity (%)
Pancreatic disease: pancreatic cancer detection					
Chu *et al.*[13]	2019	PDAC *vs* normal pancreas	CT	380	100/98.5
Liu *et al.*[14]	2020	PDAC *vs* normal pancreas	CT	690	97.3–99.0/98.9–100
Tonozuka *et al.*[15]	2020	PDAC *vs* CP *vs* normal pancreas	EUS	139	92.4/84.1
Pancreatic disease: lesion characterization: inflammatory *vs* neoplastic process					
Marya *et al.*[16]	2020	AIP *vs* normal pancreas; AIP *vs* CP; AIP *vs* PDAC	EUS	583	99/98; 94/71; 90/93
Ren *et al.*[17]	2019	MFP *vs* PDAC	CT	109	94/92
Ren *et al.*[18]	2020	MFP *vs* PDAC	Non-contrast CT	109	82.6/80.8
Park *et al.*[19]	2020	AIP *vs* PDAC	CT	182	89.7/100
Zhang *et al.*[20]	2019	AIP *vs* PDAC	^{18}F-FDG PET/CT	111	86/84
Frøkjær *et al.*[21]	2020	CP *vs* control	MRI	99	97.4/100
Pancreatic disease: prediction of prognosis and treatment response					
Xie *et al.*[22]	2020	Prediction of survival after resection of PDAC	CT	220	2.556/3.741 (hazard ratio of disease-free/overall survival)
Chakraborty *et al.*[23]	2017	2-year survival with neoadjuvant chemotherapy for resectable PDAC	CT	35	67/95
Attiyeh *et al.*[24]	2018	Prediction of survival after resection of PDAC with CA 19-9 and imaging features	CT	161	0.73 (0.68–0.78) (concordance index)
Pancreatic disease: pancreatic neuroendocrine tumor					
Canellas *et al.*[29]	2018	G1 *vs* G2/3 PNETs	CT	101	OR of texture feature entropy 3.7
Guo *et al.*[30]	2019	G3 *vs* G1/2 PNETs	CT	37	94/92
Liang *et al.*[32]	2019	G1 *vs* G2/3 PNETs	CT	137	AUC 0.891 (0.772–0.961)
Gu *et al.*[33]	2019	G1 *vs* G2/3 PNETs	CT	138	AUC 0.902 (0.798–1.0)
Bian *et al*[34]	2020	G1 *vs* G2/3 PNETs	MRI	139	AUC 0.729 (0.568–0.890)
Pancreatic disease: pancreatic cystic lesion					
Hanania *et al.*[35]	2016	Low-grade *vs* high-grade IPMN	CT	53	97/88
Chakraborty *et al.*[36]	2018	Low-risk *vs* high-risk IPMN	CT	103	84/70
Kuwahara *et al.*[37]	2019	Probability of malignancy in IPMN	EUS	206	95.7/92.6
Corral *et al.*[38]	2019	Normal *vs* low-risk IPMN *vs* high-risk IPMN/adenocarcinoma	MRI	139	75/78
Yang *et al.*[39]	2019	Serous cystadenoma *vs* mucinous cystadenoma	CT	53	85/83 (0.75)
Biliary disease: lesion characterization					
Zhang *et al.*[40]	2020	ICC *vs* CHC	CT	189	78.4/70.0 (radiomics alone)
Liu *et al.*[41]	2020	CHC *vs* non-CHC	MRI, CT	85	65/81
Biliary disease: lymph node metastasis prediction					
Ji *et al.*[42]	2019	LNM of ICC	CT	155	86.8/73.6
Ji *et al.*[43]	2019	LNM of BTC	CT	247	72.0/76.2
Yao *et al.*[44]	2020	DD of ECC; LNM of ECC	MRI	110	78.1/81.5; 83.2/79.6
Biliary disease: prognosis and treatment response evaluation					
Liang *et al.*[45]	2018	ER of ICC	MRI	209	81.0/82.0
Zhao *et al.*[46]	2019	ER of IMCC	MRI	47	93.8/83.9

(Continues)

Table 1 (Continued)

Author	Year	Clinical needs addressed	Data type	Case number	Sensitivity/specificity (%)
Mosconi *et al.*[47]	2020	Prediction of survival after TARE for ICC	CT	55	Hazard ratio of disease-free/overall survival for favorable radiomic signature 0.46 (0.22–0.95)/0.52 (0.21–1.26)
Non-imaging applications					
Jovanovic *et al.*[48]	2014	Prediction of choledocholithisis	Clinical variables	291	92.7/68.4
Tranter-Entwistle *et al.*[49]	2020	Prediction of CBD stones	Clinical variables	1315	37/96
Han *et al.*[50]	2020	Occurrence of POPF	38 clinical and surgical variables	1769	AUC 0.74
Jeong *et al.*[51]	2020	Risk of recurrence after surgical resection of ICC	12 pathological, serologic, and etiologic factors	1421	Hazard ratio of disease-free/overall survival for predicted high-risk group 3.56 (2.50–5.07)/3.19 (2.15–4.73)

AIP, autoimmune pancreatitis; BTC, biliary tract cancer; CBD, common bile duct; CHC, combined hepatocellular and cholangiocarcinoma; CP, chronic pancreatitis; CT, computed tomography; DD, differentiation degree; ER, early recurrence; G1, low grade; G2, intermediate grade; G3, high grade; ICC, intrahepatic cholangiocarcinoma; BTC, biliary tract cancer; IMCC, intrahepatic mass-forming cholangiocarcinoma; IPMN, intraductal papillary mucinous neoplasm; LNM, lymph node metastasis; MFP, mass-forming pancreatitis; MRI, magnetic resonance imaging; OR, odds ratio; PDAC, pancreatic ductal adenocarcinoma; PET, positron emission tomography; PNETs, pancreatic neuroendocrine tumors; POPF, postoperative pancreatic fistula; TARE, transarterial radioembolization.

the outcome (e.g. with or without disease) by aggregating the predictions generated by all the decision trees and tends to achieve superior accuracy over individual trees. The most frequently used method in machine learning is summarized in Figure 1.

Deep learning

Deep learning is emerging as the major AI methodology used for computer-aided detection/diagnosis.[3,9] The major advantage of DL is the ability to spontaneously capture the differential features on medical images without the requirement for explicit rule-based programming (i.e. learning from examples). The deep neural network consists of a bunch of connected nodes, each of which simulates a neuron. Each node processes information input from previous nodes and outputs the processed information to the next node, resembling the neural network of the brain. A typical DL model consists of various layers with respective purposes (Fig. 2). For example, a convolutional layer captures features from subregions within an image, and a pooling layer removes unimportant information. The predictions of the neural network are compared with the real outcomes (e.g. cancerous or noncancerous), and the methods by which the "neurons" process the information are then iteratively updated with the training data to reduce errors of model prediction. Several DL model structures including DenseNet, ResNet, and U-Net have been widely used in medical image classification, detection, or segmentation.[10–12]

Pancreatic diseases

Pancreatic cancer detection. A few studies have evaluated the performance of AI for detection of pancreatic ductal adenocarcinoma (PDAC). Chu *et al*. analyzed CT radiomic features with a machine learning algorithm (random forests) to distinguish PDAC from normal pancreas.[13] 190 patients with PDAC and 190 healthy potential renal donors as normal controls were assessed, and 40 radiomic features were selected among 478 features extracted from the pancreas for building a classification model, which achieved 99.2% accuracy (area under the curve [AUC] 0.999). Those results suggested that radiomic analysis might accurately detect PDAC. However, the radiomic features were extracted from the whole pancreas including the tumor and non-tumorous parts. It is not clear if the performance of the model could be influenced by the relative size of the tumor to the pancreas.

More recently, our group trained and tested a convolutional neural network (CNN) to distinguish PDAC and normal pancreas in CT images of 370 PDAC patients and 320 normal controls from Taiwan.[14] In the local (Taiwanese) test sets, the CNN-based analysis achieved 98.6–98.9% accuracy (AUC 0.997–0.999). Notably, in the local test sets, CNN-based analysis achieved a higher sensitivity compared with radiologist interpretation (98.3% *vs* 92.9%, difference 5.4% [95% confidence interval 1.1–9.8%]; $P = 0.014$). Notably, CNN-based analysis achieved 92.1% sensitivity for PDACs smaller than 2 cm and correctly classified 92% of PDACs missed by radiologists. When tested with data from the USA, CNN-based analysis achieved a sensitivity of 79.0%, specificity of 97.6%, and accuracy of 83.2 % (AUC 92.0%). This study provided a proof of the concept that AI may supplement

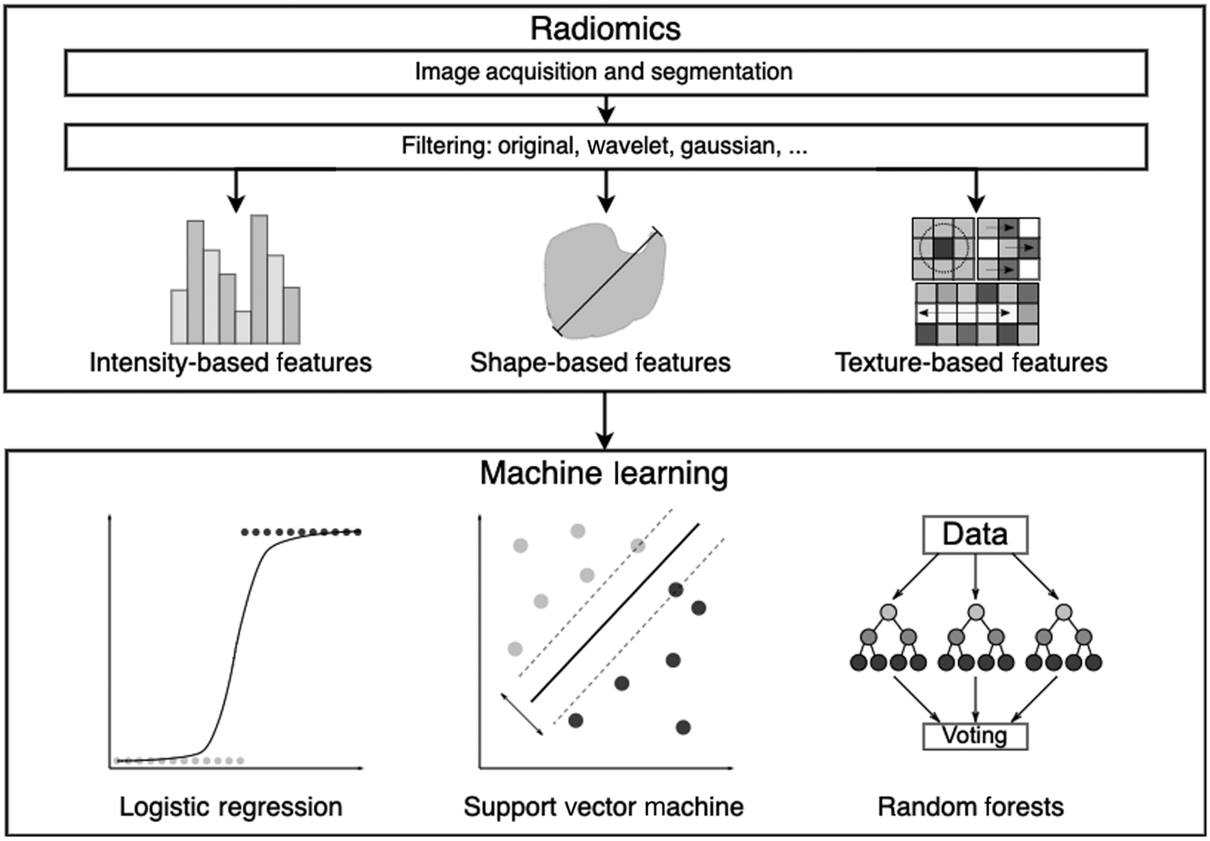

Figure 1 Radiomics and machine learning.

radiologist interpretation to facilitate early detection of PDACs. CNN has also been used for analyzing endoscopic ultrasound (EUS) images for detection of PDAC. Tonozuka *et al*. recently demonstrated that a CNN trained with EUS images of PDAC, chronic pancreatitis (CP), and normal pancreas could accurately detect PDAC in a test set (25 PDAC, 12 CP, and 10 normal) with 86.8% positive predictive value and 90.7% negative predictive value.[15] Prospective validation of the performance of AI-assisted analysis of EUS images and exploration of its potential for supplementing endoscopist interpretation and EUS-guided tissue acquisition are warranted.

Lesion characterization. Mass-forming pancreatitis (MFP) resulting from etiologies including CP and autoimmune pancreatitis (AIP) presents as a mass or focal enlargement of the pancreas on imaging. Differentiation between MFP and malignancies such as PDAC is challenging even for experienced clinicians. Accurate diagnosis is needed to avoid delayed therapy or unnecessary treatments. Marya *et al*. recently reported that a CNN model could enhance the diagnosis of AIP.[16] The CNN achieved 90% sensitivity and 93% specificity for differentiation between AIP from PDAC and 90% sensitivity and 85% specificity for distinguishing AIP from other conditions (PDAC, CP, and normal pancreas). Ren *et al*. reported that a model based on imaging features and texture in arterial-phase and portal-phase CT images could accurately differentiate MFP patients from PDAC patients (AUC 0.92–1.0).[17] The same group also reported the usefulness of radiomic features in differentiation between MFP and PDAC on non-contrast CT images with 93.3% accuracy.[18] Park *et al*. used radiomic analysis with random forest model to distinguish 89 AIP patients from 93 PDAC patients with 95.2% accuracy (AUC 97.5%).[19] Zhang *et al*. used SVM-recursive feature elimination feature selection strategy and linear SVM classifier to distinguish 45 AIP patients and 66 PDAC patients on ^{18}F-fluorodeoxyglucose PET/CT and achieved 85.0% accuracy (AUC 93.0%), outperforming radiologists and predictions based on clinical parameters.[20] Besides CT and PET/CT, radiomic analysis of MRI texture features has been shown by Frøkjær *et al*. to differentiate between CP and healthy controls with 98.0% accuracy.[21] Taken together, these studies support for the potential of AI to supplement human interpretation of EUS and radiologic images. While most of the studies focused on radiomic analysis, DL models have also shown promise and warrants further study.

Prediction of prognosis and treatment response. The current treatment options for PDAC include surgery, chemotherapy, and radiation therapy. Surgery is the only potentially curative treatment, and radiomic analysis has been used for prognosis prediction after surgical resection of PDAC.

Xie *et al*. extracted 300 radiomic features from CT images of 220 patients undergoing resection of PDAC and showed that the score calculated by radiomic model was significantly associated

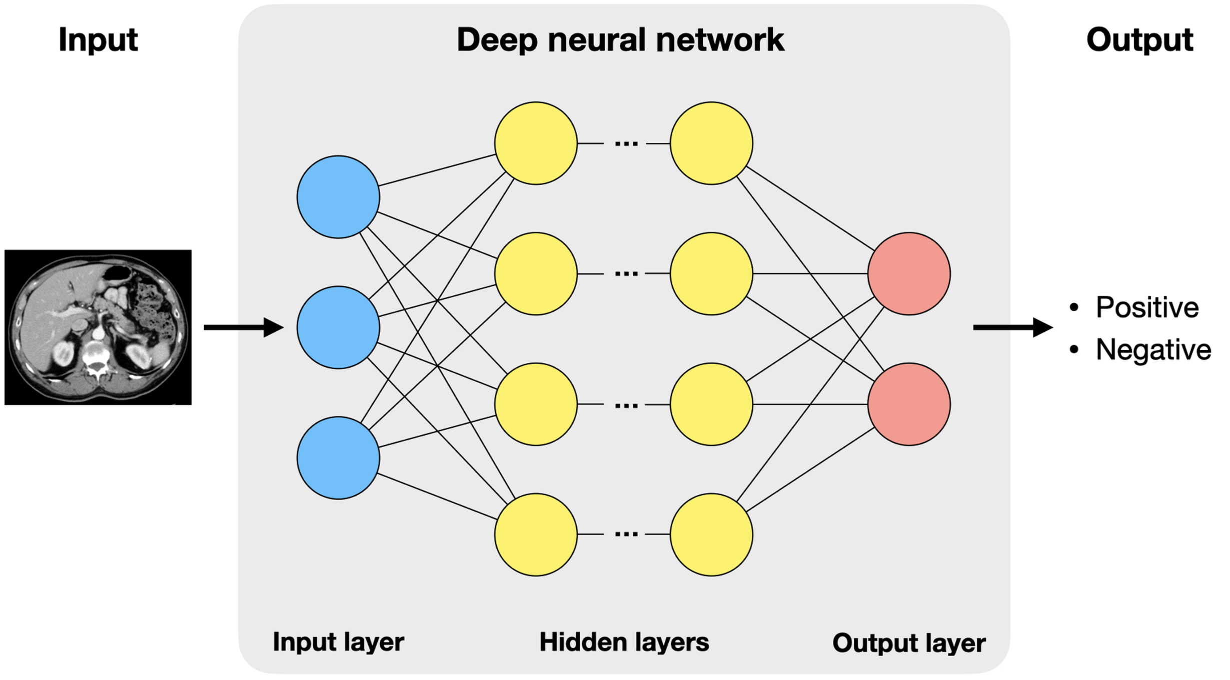

Figure 2 Deep learning. [Color figure can be viewed at wileyonlinelibrary.com]

with disease-free survival and overall survival after surgery.[22] A radiomic nomogram was developed by combining Rad-score and clinical parameters using least absolute shrinkage and selection operator regression and achieved better performance compared with predictions based on clinical parameters and TNM staging. Previous studies have shown that PDACs with high heterogeneity and low attenuation on CT were associated with worse overall survival and lower disease-free survival after resection.[23,24] As some radiomic features represent the level of attenuation and heterogeneity and thus might be useful for prognosis prediction, Yun *et al*. analyzed 88 PDAC patients receiving curative resection with radiomics and multivariate Cox regression analysis.[25] Interestingly, the results showed that homogenous texture of the tumor was associated with poorer disease-free survival, implying that homogenous texture predicted a more aggressive tumor nature. The discordant results indicate that the association between tumor attenuation/heterogeneity and postoperative outcomes of PDAC needs further investigation and suggest that radiomic profiling might be subject to influences by various factors such as scanning equipment and parameters; therefore, the results of radiomic analysis should be validated and cautiously interpreted.[26]

Pancreatic neuroendocrine tumor. Pancreatic neuroendocrine tumors (PNETs) account for 2–3% of all pancreatic tumors and are classified as low grade (G1), intermediate grade (G2), and high grade (G3) according to the Ki-67 proliferative index and mitotic counts.[27,28] Accurate assessment of tumor grade is crucial for further treatment selection and prognosis prediction. Several recent studies analyzed preoperative images with radiomic analysis for predicting tumor grade of PNET. Canellas *et al*. analyzed the CT images of 101 patients with PNETs (63 G1, 35 G2, and 3 G3 cases) before operation and showed that traditional CT features (vascular involvement, pancreatic duct dilatation, and lymphadenopathy) and texture feature (entropy) predicted tumor aggressiveness and could identify patients at risk of early disease progression after surgical resection.[29] Guo *et al*. assessed contrast-enhanced CT images of 37 PNET patients (13 G1, 11 G2, and 13 G3).[30] The results showed that arterial enhancement ratio and portal enhancement ratio combined achieved high sensitivity and specificity for differentiating G3 from G1/G2.

The reproducibility of radiomic features is influenced by multiple factors, including intra-individual test–retest repeatability, image acquisition protocol, multi-machine reproducibility, and image reconstruction parameters.[31] The generalizability of radiomic features and associated predictive models between different datasets remains a limitation. Therefore, inclusion of data from multiple sources is needed for developing robust algorithms useful in real clinical practice. Liang *et al*. conducted radiomic analysis of the arterial-phase contrast-enhanced CT images of 137 patients preoperatively to differentiate between grade 1 and grade 2/3 tumors[32] and showed that a nomogram model combining radiomic signature with clinical features achieved an AUC of 0.891 in the validation set and correlated with Ki-67 index. Gu *et al*. (training set = 104; validation set = 34) assessed arterial and portal venous-phase contrast-enhanced CT images of 138 patients for preoperative differentiation between grade 1 and grade 2/3 tumors[33] and showed that a radiomic signature significantly associated with histological grade. A nomogram combining clinical risk factors, tumor margin, and the radiomic signature achieved an AUC of 0.902 in the validation set with good calibration.

While most studies assessed CT radiomic features, MRI images can also be used for radiomic feature extraction and subsequent prediction of tumor grade prediction in PNET. Bian *et al*. assessed the radiomic features of non-contrast MRI images from 139 PNET patients for differentiation between G1 and G2/3 tumors,[34] and a model that combined the radiomic signature and 14 imaging features achieved an AUC of 0.729 in the validation set.

Pancreatic cystic lesions. Intraductal papillary mucinous neoplasms (IPMNs) represent 15–30% of cystic lesions of the pancreas and are a precursor lesion to PDAC. Radiomic analysis has been studied for predicting the risk of malignant transformation. Hanania *et al*. studied 53 IPMN patients who subsequently underwent surgical resection (34 high grade and 19 low grade) and showed that 14 radiomic features of the gray-level co-occurrence matrix category predicted the histopathological grade with an AUC of 0.96.[35] Chakraborty *et al*. analyzed 103 patents with branch-duct IPMNs for discriminating between low-risk and high-risk IPMN.[36] The prediction model based on radiomic features achieved an AUC of 0.77, and the model incorporating those features and clinical variables achieved an AUC of 0.81. Alternatively, Kuwahara *et al*. analyzed EUS images of IPMNs for training a DL model for predicting the probability of malignancy in IPMNs.[37] The DL model achieved a higher accuracy compared with human diagnosis and the presence of mural nodules (94.0%, 56.0%, and 68.0%, respectively).

Deep learning has also been applied in predicting the malignancy risk of IPMN. Corral *et al*. analyzed T2-weighted and post-contrast T1-weighted MRI images of 139 patients consisting of normal pancreas (20%), IPMN with low-grade dysplasia (34%), IPMN with high-grade dysplasia (14%), and adenocarcinoma (29%).[38] The trained DL model achieved comparable performance (AUC 0.78) to current predictive criteria (AUCs 0.76 for American Gastroenterology Association criteria and 0.77 for Fukuoka criteria).

Besides risk stratification for IPMN, radiomics and machine learning methods have also been used in other cystic lesions. Yang *et al*. analyzed 53 patients with pancreatic serous cystadenoma and 25 patients with pancreatic mucinous cystadenoma.[39] Textural parameters of the cystic neoplasms were extracted from contrast-enhanced CT and analyzed using random forest, and the model could discriminate pancreatic mucinous cystadenomas from serous cystadenomas with acceptable performance (AUC 0.75). Collectively, those studies suggest that analysis of preoperative imaging using either radiomic analysis or DL might represent a potential computer-aided detection/diagnosis tool for risk stratification in pancreatic cystic lesions, but further studies with larger sample sizes are needed to validate those findings.

Biliary diseases

Current AI research in biliary diseases mainly focuses on oncologic applications, and cholangiocarcinoma is the most widely studied. Radiomic analysis is the most widely employed method and can be used for lesion characterization, prediction of lymph node metastasis and prognosis, and treatment response evaluation.

Lesion characterization. Zhang *et al*. developed a radiomic-based model from contrast-enhanced CT images of 86 patients with intrahepatic cholangiocarcinoma (ICC) and 46 patents with combined hepatocellular and cholangiocarcinoma (CHC).[40] The radiomic features extracted from tumors in the arterial phase were selected to build a radiomic score, which yielded an AUC of 0.789 for differentiation between ICC and CHC in the validation cohort, and a model including Rad-score and four clinical–radiologic factors achieved an AUC of 0.942. Liu *et al*. analyzed CT and MRI radiomic features with support vector machine for classifying hepatic malignancies.[41] Prediction based on MRI radiomic features achieved the best performance in differentiating CHC from non-CHC (AUC 0.77).

Prediction of lymph node metastasis. Ji *et al*. developed a radiomic model from ICC patients who underwent curative resection and lymphadenectomy.[42] Radiomic features were extracted from arterial-phase CT images, and the radiomic signature was built with the least absolute shrinkage and selection operator method, and multivariate logistic regression analysis was used to construct a model incorporating radiomic signature and other independent predictors to predict lymph node metastasis. A radiomic nomogram that incorporated radiomic signature and CA 19-9 level showed good calibration and discrimination in the validation cohort (AUC 0.89) and independently predicted overall and recurrence-free survival. The same group further developed a radiomic model for predicting lymph node metastasis using portal venous CT scans of patients with biliary cancers.[43] The radiomic signature composed of three nodal status-related features was significantly associated with lymph node metastasis (AUC 0.77). Besides CT radiomic, Yao *et al*. studied patients with extrahepatic cholangiocarcinoma by extracting radiomic features from images obtained with precontrast T1-weighted imaging, T2-weighted imaging, and diffusion-weighted imaging, and a radiomic model could predict lymph node metastasis of extrahepatic cholangiocarcinoma with an AUC of 0.89.[44] Those studies suggest that analysis of CT/MRI radiomic features may assist in predicting lymph node metastasis in biliary cancers.

Prediction of prognosis and treatment response. The treatment options of biliary tract cancer included surgery, liver transplantation, chemotherapy, and radiation therapy. While resection is potentially curative, the risk of recurrence is high. Liang *et al*. developed a predictive model for early recurrence in 209 ICC patients (139 training and 70 validation) after partial hepatectomy.[45] They extracted radiomic features from contrast-enhanced MRI and constructed a radiomic signature, and the radiomic nomogram combining the radiomic signature and clinical stage achieved an AUC of 0.86 in the validation set. Zhao *et al*. similarly used four MRI sequences including fat suppression T2-weighted imaging, arterial-phase, portal venous-phase, and delayed-phase contrast-enhanced imaging for predicting recurrence of ICC.[46] The combined model including enhancement patterns, expression of vascular endothelial growth factor receptor, and radiomic features achieved better performance in predicting early recurrence compared with the radiomic model or clinicoradiologic–pathological model alone, achieving an accuracy of 87.2%. To predict treatment response, Mosconi *et al*. used radiomic analysis to quantify vascularization and homogeneity of ICC in patients receiving transarterial radioembolization.[47] The results showed that patients with a favorable radiomic signature had a significantly longer median progression-free survival (12.1 months) than that of patients with an unfavorable signature (5.1 months). These results are in line with the notion that radiomic analysis can quantitatively assess the characteristics of the tumor, which are associated with the treatment response.

Non-imaging applications of artificial intelligence. Similar with medical imaging data, AI models can also process non-imaging data to assist in tasks such as disease risk stratification and prognosis prediction. Jovanovic *et al*. trained an artificial neural network, which predicted the presence/absence of choledocholithiasis in patients based on blood tests (levels of total bilirubin, transaminases, alkaline phosphatase, and γ-glutamyl transferase), sonographic findings (diameter of common bile duct and presence/absence of intraductal hyperechoic lesions), and white blood cell count.[48] Using ERCP findings as reference, the artificial neural network was sensitive and moderately specific in predicting the presence of choledocholithiasis (AUC 0.884). Tranter-Entwistle *et al*. explored the potential usefulness of AI for predicting common bile duct stones in patients with acute biliary disease and achieved a sensitivity of 37% and a specificity of 96% based on the results of blood tests.[49] Notably, in that study, the authors described challenges posed by coding errors and missing data in real-world clinical data, highlighting the hurdles of such research and the need for large datasets to achieve adequate model performance. Collectively, those studies supported that AI-based predictive models might facilitate risk stratification and selection of patients for therapeutic ERCP. Han *et al*. reported that a neural network could predict the occurrence of postoperative pancreatic fistula with an AUC of 0.74 by incorporating variables including baseline clinical features (demographics, comorbidities, blood counts, and biochemistry including amylase and lipase levels) and surgical parameters (portal vein resection and anastomotic methods).[50] Jeong *et al*. used a DL neural network to predict the risk of recurrence after surgical resection of ICC based on 12 parameters. The finding showed that the AI framework performed better compared with the American Joint Committee on Cancer stage and Cox regression analysis (AUC 0.78 *vs* 0.60 and 0.70).[51] Springer *et al*. reported prediction of the malignant potential (little/no, small, and high) of pancreatic cystic lesions based on supervised machine learning-based analyses of clinical features, imaging characteristics, and cyst fluid genetic and biochemical analyses and showed that AI-based predictions were more accurate than conventional clinical and imaging criteria alone in guiding treatment selection and would reduce unnecessary cyst resections.[52] Those studies highlight the potential advantages of AI over traditional statistical methods to integrate data of various types/multi-omics data for prognosis prediction and treatment selection.

Conclusion

The current applications of AI in pancreatobiliary diseases focus mainly on imaging studies to assist in tumor detection, lesion characterization, and prognosis prediction. Compared with radiomic analysis with machine learning that has been extensively evaluated for applications in pancreatobiliary diseases, research evaluating the usefulness of DL is relatively limited but suggests promising potentials. Overall, most studies are of modest sample size and need to be validated with respect to reproducibility and generalizability across institutions and populations. The potential usefulness of AI in supplementing clinicians for caring patients with pancreatobiliary diseases needs to be explored in greater width and depth, and their potential impact in patient care should be further examined.

References

1 Jiang F, Jiang Y, Zhi H *et al*. Artificial intelligence in healthcare: past, present and future. *Stroke Vasc. Neurol.* 2017; **2**: 230–43. https://doi.org/10.1136/svn-2017-000101

2 Yasaka K, Abe O. Deep learning and artificial intelligence in radiology: current applications and future directions. *PLoS Med.* 2018; **15**: e1002707. https://doi.org/10.1371/journal.pmed.1002707

3 Erickson BJ, Korfiatis P, Akkus Z, Kline TL. Machine learning for medical imaging. *Radiographics* 2017; **37**: 505–15. https://doi.org/10.1148/rg.2017160130

4 Gillies RJ, Kinahan PE, HH. Radiomics: images are more than pictures, they are data. *Radiology* 2016; **278**: 563–77. https://doi.org/10.1148/radiol.2015151169

5 Chartrand G, Cheng PM, Vorontsov E *et al*. Deep learning: a primer for radiologists. *Radiographics* 2017; **37**: 2113–31. https://doi.org/10.1148/rg.2017170077

6 Kumar V, Gu Y, Basu S *et al*. Radiomics: the process and the challenges. *Magn. Reson. Imaging* 2012; **30**: 1234–48. https://doi.org/10.1016/j.mri.2012.06.010

7 Cortes C, Vapnik V. Support-vector networks. *Mach. Learn* 1995; **20**: 273–97. https://doi.org/10.1023/A:1022627411411

8 Breiman L. Random forests. *Mach. Learn* 2001; **45**: 5–32. https://doi.org/10.1023/A:1010933404324

9 Litjens G, Kooi T, Bejnordi BE *et al*. A survey on deep learning in medical image analysis. *Med. Image Anal.* 2017; **42**: 60–88. https://doi.org/10.1016/j.media.2017.07.005

10 Huang G, Liu Z, van der Maaten L, Weinberger KQ. Densely connected convolutional networks [internet]. *arXiv [cs.CV]* 2016. http://arxiv.org/abs/1608.06993

11 He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition [Internet]. *arXiv [cs.CV]* 2015. Available at: http://arxiv.org/abs/1512.03385

12 Ronneberger O, Fischer P, Brox T. U-Net: Convolutional networks for biomedical image segmentation. In: *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015*. Springer International Publishing, 2015; 234–41.Available at: https://doi.org/10.1007/978-3-319-24574-4_28

13 Chu LC, Park S, Kawamoto S *et al*. Utility of CT radiomics features in differentiation of pancreatic ductal adenocarcinoma from normal pancreatic tissue. *AJR Am. J. Roentgenol.* 2019: 1–9. https://doi.org/10.2214/AJR.18.20901

14 Liu K-L, Wu T, Chen P-T *et al*. Deep learning to distinguish pancreatic cancer tissue from non-cancerous pancreatic tissue: a retrospective study with cross-racial external validation. *Lancet Dig. Health* 2020; **2**: e303–13.

15 Tonozuka R, Itoi T, Nagata N *et al*. Deep learning analysis for the detection of pancreatic cancer on endosonographic images: a pilot study. *J. Hepatobiliary Pancreat. Sci.* 2020; https://doi.org/10.1002/jhbp.825

16 Marya NB, Powers PD, Chari ST *et al*. Utilisation of artificial intelligence for the development of an EUS-convolutional neural network model trained to enhance the diagnosis of autoimmune pancreatitis. *Gut* 2020 [cited 9 December 2020];

Available at: https://gut.bmj.com/content/early/2020/10/07/gutjnl-2020-322821
17 Ren S, Zhang J, Chen J *et al*. Evaluation of texture analysis for the differential diagnosis of mass-forming pancreatitis from pancreatic ductal adenocarcinoma on contrast-enhanced CT images. *Front. Oncol.* 2019; **9**: 1171. https://doi.org/10.3389/fonc.2019.01171
18 Ren S, Zhao R, Zhang J *et al*. Diagnostic accuracy of unenhanced CT texture analysis to differentiate mass-forming pancreatitis from pancreatic ductal adenocarcinoma. *Abdom. Radiol. (NY)* 2020; **45**: 1524–33. https://doi.org/10.1007/s00261-020-02506-6
19 Park S, Chu LC, Hruban RH *et al*. Differentiating autoimmune pancreatitis from pancreatic ductal adenocarcinoma with CT radiomics features. *Diagn. Interv. Imaging* 2020; **101**: 555–64. https://doi.org/10.1016/j.diii.2020.03.002
20 Zhang Y, Cheng C, Liu Z *et al*. Radiomics analysis for the differentiation of autoimmune pancreatitis and pancreatic ductal adenocarcinoma in ^{18}F-FDG PET/CT. *Med. Phys.* 2019; **46**: 4520–30. https://doi.org/10.1002/mp.13733
21 Frøkjær JB, Lisitskaya MV, Jørgensen AS *et al*. Pancreatic magnetic resonance imaging texture analysis in chronic pancreatitis: a feasibility and validation study. *Abdom. Radiol. (NY)* 2020; **45**: 1497–506. https://doi.org/10.1007/s00261-020-02512-8
22 Xie T, Wang X, Li M, Tong T, Yu X, Zhou Z. Pancreatic ductal adenocarcinoma: a radiomics nomogram outperforms clinical model and TNM staging for survival estimation after curative resection. *Eur. Radiol.* 2020; **30**: 2513–24. https://doi.org/10.1007/s00330-019-06600-2
23 Chakraborty J, Langdon-Embry L, Cunanan KM *et al*. Preliminary study of tumor heterogeneity in imaging predicts two year survival in pancreatic cancer patients. *PLoS One* 2017; **12**: e0188022. https://doi.org/10.1371/journal.pone.0188022
24 Attiyeh MA, Chakraborty J, Doussot A *et al*. Survival prediction in pancreatic ductal adenocarcinoma by quantitative computed tomography image analysis. *Ann. Surg. Oncol.* 2018; **25**: 1034–42. https://doi.org/10.1245/s10434-017-6323-3
25 Yun G, Kim YH, Lee YJ, Kim B, Hwang J-H, Choi DJ. Tumor heterogeneity of pancreas head cancer assessed by CT texture analysis: association with survival outcomes after curative resection. *Sci. Rep.* 2018; **8**: 7226. https://doi.org/10.1038/s41598-018-25627-x
26 Rizzo S, Botta F, Raimondi S *et al*. Radiomics: the facts and the challenges of image analysis. *Eur. Radiol. Exp.* 2018; **2**: 36. https://doi.org/10.1186/s41747-018-0068-z
27 Fraenkel M, Kim MK, Faggiano A, Valk GD. Epidemiology of gastroenteropancreatic neuroendocrine tumours. *Best Pract. Res. Clin. Gastroenterol.* 2012; **26**: 691–703. https://doi.org/10.1016/j.bpg.2013.01.006
28 Guilmette JM, Nosé V. Neoplasms of the neuroendocrine pancreas: an update in the classification, definition, and molecular genetic advances. *Adv Anat Pathol* 2019; **26**: 13–30. https://doi.org/10.1097/PAP.0000000000000201
29 Canellas R, Burk KS, Parakh A, Sahani DV. Prediction of pancreatic neuroendocrine tumor grade based on CT features and texture analysis. *AJR Am. J. Roentgenol.* 2018; **210**: 341–6. https://doi.org/10.2214/AJR.17.18417
30 Guo C, Zhuge X, Wang Z *et al*. Textural analysis on contrast-enhanced CT in pancreatic neuroendocrine neoplasms: association with WHO grade. *Abdom. Radiol. (NY)* 2019; **44**: 576–85. https://doi.org/10.1007/s00261-018-1763-1
31 Park JE, Park SY, Kim HJ, Kim HS. Reproducibility and generalizability in radiomics modeling: possible strategies in radiologic and statistical perspectives. *Korean J. Radiol.* 2019; **20**: 1124–37. https://doi.org/10.3348/kjr.2018.0070
32 Liang W, Yang P, Huang R *et al*. A combined nomogram model to preoperatively predict histologic grade in pancreatic neuroendocrine tumors. *Clin. Cancer Res.* 2019; **25**: 584–94. https://doi.org/10.1158/1078-0432.CCR-18-1305
33 Gu D, Hu Y, Ding H *et al*. CT radiomics may predict the grade of pancreatic neuroendocrine tumors: a multicenter study. *Eur. Radiol.* 2019; **29**: 6880–90. https://doi.org/10.1007/s00330-019-06176-x
34 Bian Y, Zhao Z, Jiang H *et al*. Noncontrast radiomics approach for predicting grades of nonfunctional pancreatic neuroendocrine tumors. *J. Magn. Reson. Imaging* 2020; **52**: 1124–36. https://doi.org/10.1002/jmri.27176
35 Hanania AN, Bantis LE, Feng Z, Wang H, Tamm EP. Quantitative imaging to evaluate malignant potential of IPMNs. *Oncotarget* 2016; https://www.ncbi.nlm.nih.gov/pmc/articles/pmc5349873/
36 Chakraborty J, Midya A, Gazit L *et al*. CT radiomics to predict high-risk intraductal papillary mucinous neoplasms of the pancreas. *Med. Phys.* 2018; **45**: 5019–29. https://doi.org/10.1002/mp.13159
37 Kuwahara T, Hara K, Mizuno N *et al*. Usefulness of deep learning analysis for the diagnosis of malignancy in intraductal papillary mucinous neoplasms of the pancreas. *Clin. Transl. Gastroenterol.* 2019; **10**: 1–8. https://doi.org/10.14309/ctg.0000000000000045
38 Corral JE, Hussein S, Kandel P, Bolan CW, Bagci U, Wallace MB. Deep learning to classify intraductal papillary mucinous neoplasms using magnetic resonance imaging. *Pancreas* 2019; **48**: 805–10. https://doi.org/10.1097/MPA.0000000000001327
39 Yang J, Guo X, Ou X, Zhang W, Ma X. Discrimination of pancreatic serous cystadenomas from mucinous cystadenomas with CT textural features: based on machine learning. *Front. Oncol.* 2019; **9**:494. https://doi.org/10.3389/fonc.2019.00494
40 Zhang J, Huang Z, Cao L, *et al*. Differentiation combined hepatocellular and cholangiocarcinoma from intrahepatic cholangiocarcinoma based on radiomics machine learning. *Ann. Transl. Med.* 2020; **8**(4):119. https://doi.org/10.21037/atm.2020.01.126
41 Liu X, Khalvati F, Namdar K *et al*. Can machine learning radiomics provide pre-operative differentiation of combined hepatocellular cholangiocarcinoma from hepatocellular carcinoma and cholangiocarcinoma to inform optimal treatment planning? *Eur. Radiol.* 2020; https://doi.org/10.1007/s00330-020-07119-7
42 Ji G-W, Zhu F-P, Zhang Y-D *et al*. A radiomics approach to predict lymph node metastasis and clinical outcome of intrahepatic cholangiocarcinoma. *Eur. Radiol.* 2019; **29**: 3725–35. https://doi.org/10.1007/s00330-019-06142-7
43 Ji G-W, Zhang Y-D, Zhang H *et al*. Biliary tract cancer at CT: a radiomics-based model to predict lymph node metastasis and survival outcomes. *Radiology* 2019; **290**: 90–8. https://doi.org/10.1148/radiol.2018181408
44 Yao X, Huang X, Yang C *et al*. A novel approach to assessing differentiation degree and lymph node metastasis of extrahepatic cholangiocarcinoma: prediction using a radiomics-based particle swarm optimization and support vector machine model. *JMIR Med. Inform.* 2020; **8**: e23578. https://doi.org/10.2196/23578
45 Liang W, Xu L, Yang P *et al*. Novel nomogram for preoperative prediction of early recurrence in intrahepatic cholangiocarcinoma. *Front. Oncol.* 2018; **8**: 360. https://doi.org/10.3389/fonc.2018.00360
46 Zhao L, Ma X, Liang M *et al*. Prediction for early recurrence of intrahepatic mass-forming cholangiocarcinoma: quantitative magnetic resonance imaging combined with prognostic immunohistochemical markers. *Cancer Imaging* 2019; **19**: 49.
47 Mosconi C, Cucchetti A, Bruno A *et al*. Radiomics of cholangiocarcinoma on pretreatment CT can identify patients who would best respond to radioembolisation. *Eur. Radiol.* 2020. https://doi.org/10.1007/s00330-020-06795-9
48 Jovanovic P, Salkic NN, Zerem E. Artificial neural network predicts the need for therapeutic ERCP in patients with suspected choledocholithiasis. *Gastrointest. Endosc.* 2014; **80**: 260–8.

49 Tranter-Entwistle I, Wang H, Daly K, Maxwell S, Connor S. The challenges of implementing artificial intelligence into surgical practice. *World J. Surg.* 2020; https://doi.org/10.1007/s00268-020-05820-8
50 Han IW, Cho K, Ryu Y *et al*. Risk prediction platform for pancreatic fistula after pancreatoduodenectomy using artificial intelligence. *World J. Gastroenterol.* 2020; **26**: 4453–64. https://doi.org/10.3748/wjg.v26.i30.4453
51 Jeong S, Ge Y, Chen J *et al*. Latent risk intrahepatic cholangiocarcinoma susceptible to adjuvant treatment after resection: a clinical deep learning approach. *Front. Oncol.* 2020; **10**: 143. https://doi.org/10.3389/fonc.2020.00143
52 Springer S, Masica DL, Dal Molin M *et al*. A multimodality test to guide the management of patients with a pancreatic cyst. *Sci. Transl. Med.* 2019; **11**. https://doi.org/10.1126/scitranslmed.aav4772